

CONSENSUS ON RECOMMENDED FUNCTIONS OF A SMART HOME SYSTEM TO SUPPORT PEOPLE LIVING WITH DEMENTIA: A MODIFIED DELPHI STUDY

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INTRODUCTION

Dementia is characterized by a progressive decline in cognitive functioning that leads to an increased dependency on long-term care [1]. Home-based care programs can benefit people living with dementia by maintaining established routines within familiar environments and may delay admission to higher-cost residential care [2]. Smart home technologies that use interconnected devices and appliances to collect and share data via Wi-Fi, could enhance clinical care by enabling remote monitoring and automated support for people living with dementia [3]. However, despite the rapidly evolving field of smart home technologies there is limited evidence of their effectiveness in this population [4]. This study aimed to establish consensus among healthcare professionals regarding the importance of core functions for inclusion in a smart home to support people living with dementia.

METHODS

Healthcare professionals experienced in caring for people living with dementia participated in a three-stage modified DELPHI process with two survey rounds followed by an online consensus workshop. In Round 1, participants used a Likert scale (1 – not important) to 9 (very important) to rate 46 items grouped into ten domains, informed by current smart home interventions that aligned with Australian Clinical Practice Guidelines for People with Dementia [5]. Consensus was defined as $\geq 70\%$ participants rating items as

important (scores ≥ 7) and $\leq 15\%$ rating them as unimportant (≤ 3). For Round 2, participants were asked to rate items that had not reached consensus in Round 1 in addition to any new items. Free text comments and workshop discussion data was analysed to identify themes and sub-themes related to smart home functions.

RESULTS AND DISCUSSION

In total, 28 health professionals (64% female) completed the Round 1 survey with consensus for 22 functions as very important (Figure 1). Two items were modified for round two. A further 2 items reached consensus as very important in Round 2 and an additional 6 items during the workshop.

Preliminary analysis of workshop data highlighted the need for adaptable and customized smart home functions, consideration for ethical inclusion of people living with dementia; provision of technical support for implementation and a hybrid healthcare model.

CONCLUSIONS

The findings of this study can inform the future prioritization and implementation of smart home technologies to support people living with dementia.

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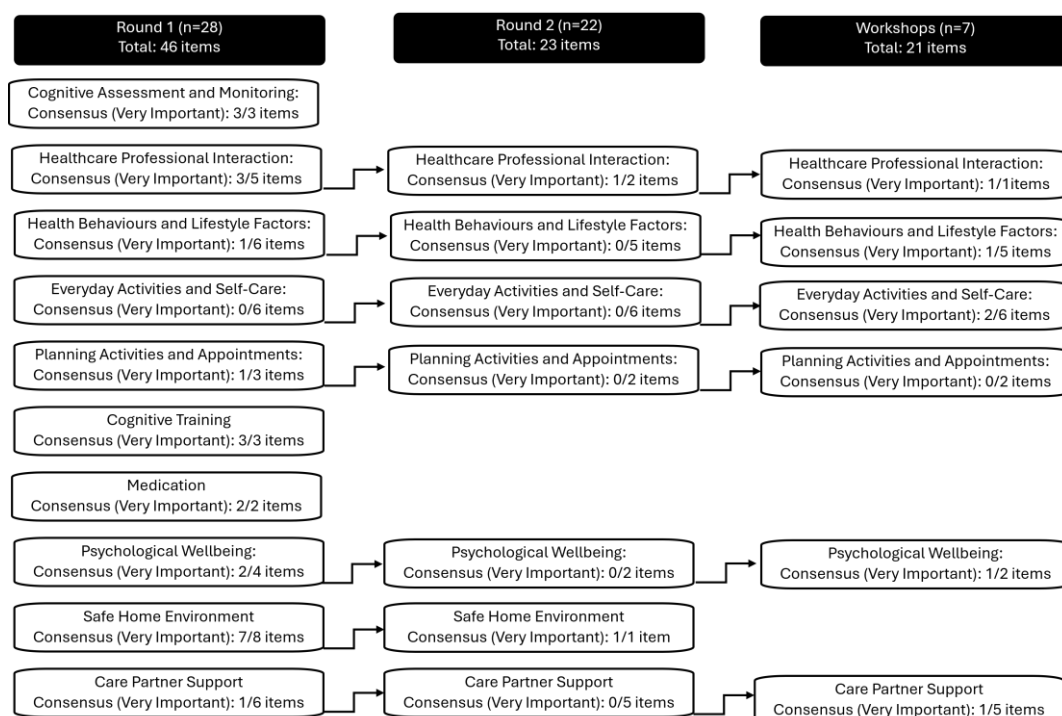


Figure 1: Flowchart of participants and consensus item outcomes by domain